Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think that I will be able to use many of the previous assignments to help me to complete this project, but especially the couple drafter assignments that we have had. I think I will be using loops, conditionals, lists, and dataclasses when programming my website. When completing this project, I want to gain a stronger understanding of drafter because that is something that I haven't understood as well during the semester.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think it won't be too hard to make the custom schedule based off of the hobby, experience level, and time given by the user. It could involve a lot of conditionals, but I think I will be able to figure that out pretty easily. I think it will be challenging to create the checklist for which parts of the practice have been completed because I think this will involve a lot of complicated code in order to be able to check off the correct part of the practice when finished with it.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think making the checklist part of the website for checking off which parts of the practice the user has completed will require the most time and effort because there is a lot of code that goes into making a checklist work. I think I should first add the home screen with buttons to take you to the screen where you make your decisions to get your practice schedule. So, I would probably make the code to get the schedule before making the code for the checklist

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

My phone is often a distraction when I am trying to get my work done because I am tempted to check it. In order to minimize distractions, I can put my phone far away from myself so I won't be as tempted to check it. While working, I will need the LLM to help me with the code and I will need the documents that I filled out for my project planning so I know what I need to be coding.

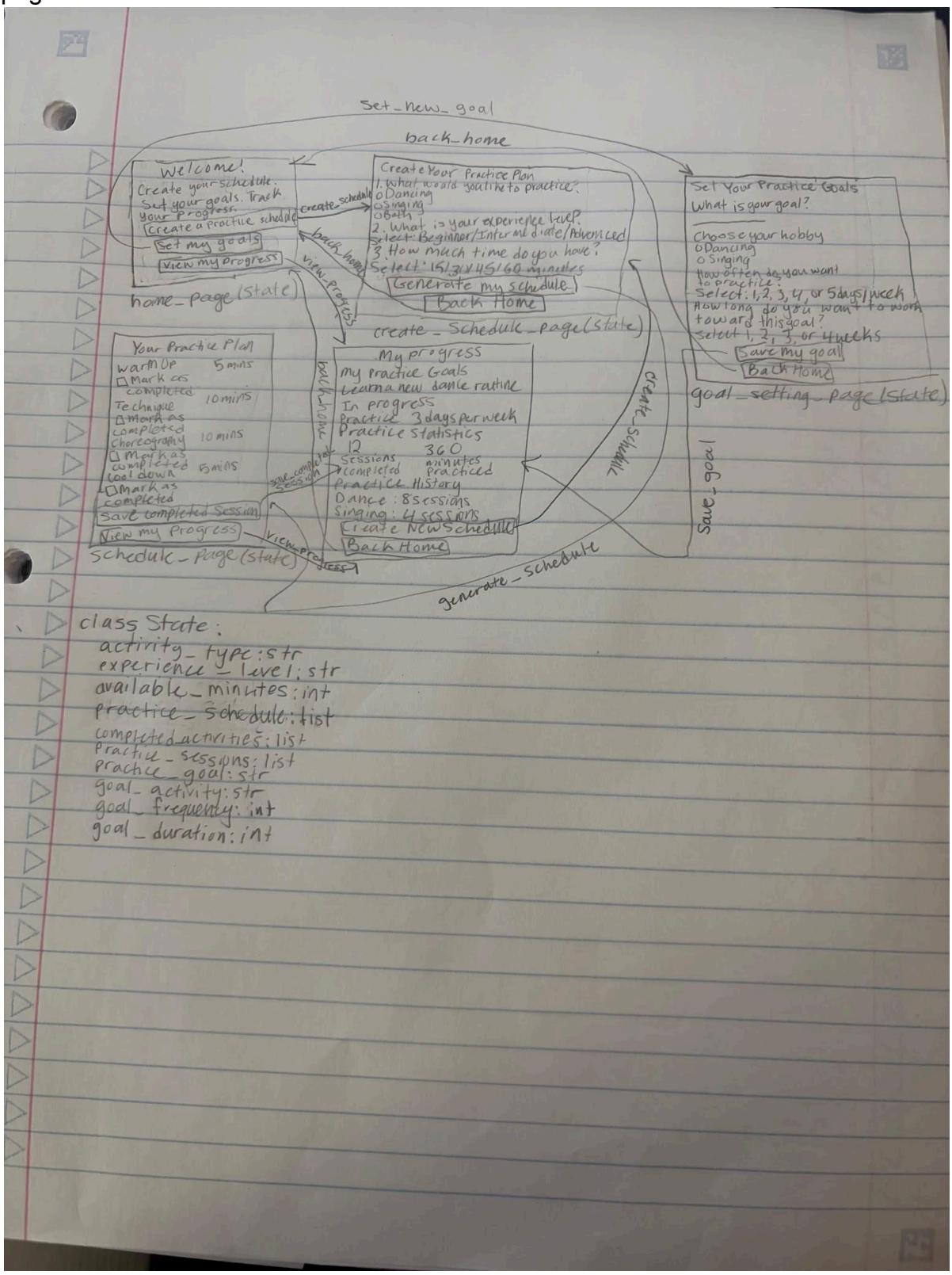
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.




```
class State:
    activity_type: str
    experience_level: str
    available_minutes: int
    practice_schedule: list
    completed_activities: list
    Practice_sessions: list
    practice_goal: str
    goal_activity: str
    goal_frequency: int
    goal_duration: int
```

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose of my website is to allow singers and dancers to create a practice schedule, set goals, and track their practice progress.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	activity_type: str - Gets modified when the user selects Dance, Singing, or Both on the create_schedule_page experience_level: str - Gets modified when the user selects Beginner, Intermediate, or Advanced on the create_schedule-page available_minutes: int - Gets modified when the user selects 15, 30, 45, or 60 minutes on the create_schedule_page practice_schedule: list - Gets modified when the program generates a schedule based on the user's preferences on the schedule_page completed_activities: list - Gets modified when the user checks off activities as they complete them on the schedule_page practice_sessions: list - Gets modified when the program add a record when the user saves a completed session on the schedule_page practice_goal: str - Gets modified when the user enters their personal improvement goal on the goal_setting_page goal_activity: str - Gets modified when the user selects whether their goal relates to dancing or singing on the goal_setting_page. goal_frequency: int - Gets modified when the user selects how many days per week they want to

	practice on the goal_setting_page goal_duration: int - Gets modified when the user selects how many weeks they want to work toward their goal on the goal_setting page.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	The second dataclass would be PracticeActivity which would have the attributes: activity_name: str - Gets modified when the app creates an activity, such as “Vocal Warm-ups” or “Choreography” activity_type: str - Gets modified when the activity is created as either Dancing or Singing difficulty: str - Gets modified based on the user’s selected experience level duration: int - Gets modified when the app assigns a number of minutes to the activity completed: bool - Gets modified when the user checks off the activity as completed and will change from False to True
Routes (minimum 7) The names of your route functions	back_home, create_schedule, set_new_goal, generate_schedule, view_progress, save_goal, save_completed_session
Pages (minimum 5) A brief description of the pages in your site	home_page - welcomes the user to the website and allows them to decide if they want to create a practice schedule, set goals, or view their progress. create_schedule_page - allows the user to create their practice schedule by selecting the options they want for their activity, experience level, available practice time, and activity to practice. They can then generate their schedule. goal_setting_page - allows the user to enter their goal, select which hobby it is for, how often they want to practice and how long they want to work toward their goal. They can then either save their goal or go back to the home page. schedule_page - allows the user to view their generated practice schedule and check off the parts

	of the practice as they complete them. They can then save their completed session or view their progress. progress_page - allows the user to view their saved goals, practice statistics, practice history. They can then create a new schedule or go back to the home page.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	There will be CheckBox for when the user checks off a part of their practice being completed. There will be TextBox for the goal setting because the user will type in the goal they want to accomplish. There will be SelectBox for when the user selects the hobby they want to practice, their experience level, how much time they have, their hobby for their goal, how often they want to practice for their goal, and how long they want to work toward their goal.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	There would be nested if statements that determine the schedule outputted to the user based on if they selected singing or dancing and then it would depend on what experience level they selected, and finally it would depend on the amount of time they selected. So, it would end up being a bunch of nested if statements to determine what the result will be for the schedule.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	When displaying all of the activities in the user's practice schedule, a loop would be used to iterate through practice_schedule which is a list in order for the website to display each activity's name, duration, and completion status.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

What would a second dataclass be for the website in addition to state?

For Rhythm & Routine, I recommend making your second dataclass `PracticeActivity`.

This would work really well alongside your `State` dataclass because your website needs to store information about individual dancing and singing activities.

1. Your second dataclass: `PracticeActivity`

Each `PracticeActivity` object represents one activity that can appear in a user's personalized practice schedule.

For example:

- Vocal warm-ups
- Choreography
- Dance technique
- Breathing exercises
- Song practice

Here are the attributes I recommend:

Attribute	Type	Purpose
`activity_name`	`str`	Name of the practice activity
`activity_type`	`str`	Dance or Singing
`difficulty`	`str`	Beginner, Intermediate, or Advanced
`duration`	`int`	How many minutes the activity takes
`completed`	`bool`	Whether the user completed the activity

2. Example of what the dataclass would look like

```
from dataclasses import dataclass
```

```
@dataclass
class PracticeActivity:
    activity_name: str
    activity_type: str
    difficulty: str
    duration: int
    completed: bool
```

You could create individual objects like this:

```
activity1 = PracticeActivity(
    "Vocal Warm-ups",
    "Singing",
    "Beginner",
    5,
    False
)


activity2 = PracticeActivity(
    "Choreography",
    "Dance",
    "Intermediate",
    15,
    False
)
```

3. How it works with State

Your `State` stores the user's information and overall website data, while `PracticeActivity` describes individual activities.

State dataclass

Stores:

- User preferences
- Practice goals
- Generated schedule
- Practice session history

PracticeActivity dataclass

Stores individual activities:

- Activity name
- Activity type
- Difficulty
- Duration
- Completion status

For example, your State could contain a list of `PracticeActivity` objects:

```
practice_schedule = [
    activity1,
    activity2
]
```

This way, your schedule can contain multiple activities, and users can mark each one as completed.

4. Why I recommend this dataclass

It connects directly to all three of your website's main features:

- Personalized schedules: Your program selects activities based on the user's preferences and experience level.
- Practice goal setting: Activities can help users work toward their selected goals.
- Completion checklist: The `completed` attribute tracks which activities users finish.

My recommendation

Use `PracticeActivity` as your second dataclass. It is simple enough for a CS1 project, uses concepts you're learning in Python, and has a clear purpose within your website.

You also don't need to create a separate dataclass for every activity. You can create as

many `PracticeActivity` objects as you need using this one class!

what would an example of a loop be in the website

A great example of a loop in Rhythm & Routine would be displaying all the activities in the user's practice schedule.

For example, if the user has three activities scheduled, a `for` loop can go through each activity and display its name, duration, and completion status.

Example: Displaying practice activities

```
for activity in state.practice_schedule:
    print(activity.activity_name)
    print(activity.duration, "minutes")


    if activity.completed:
        print("Completed!")
    else:
        print("Not completed yet")
```

How it works

1. The loop starts with the first activity in `state.practice_schedule`.
2. It displays the activity's name and duration.
3. It checks whether the activity is completed.
4. It repeats these steps for every activity in the schedule.
5. The loop ends when there are no more activities.

Example output

If the user has three activities, the output could look like this:

```
Vocal Warm-ups
5 minutes
Completed!

Dance Practice
15 minutes
Not completed yet
```

```
Stretching
5 minutes
Not completed yet
```

How you could explain this in your project

"My website uses a for loop to go through the user's practice schedule and display each activity, its duration, and whether it has been completed. This allows the website to display schedules of different lengths without having to write separate code for every activity."

Why this is useful: It makes your website more flexible because the user can have a different number of practice activities each time they create a schedule.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.